An Intelligent Two-Layer Intrusion Detection System for the Internet of Things

Mohammed M. Alani, *Senior Member, IEEE*, and Ali Ismail Awad, *Senior Member, IEEE*

***Abstract*—The Internet of Things (IoT) has become an enabler paradigm for different applications, such as healthcare, education, agriculture, smart homes, and recently, enterprise systems. Significant advances in IoT networks have been hindered by security vulnerabilities and threats, which, if not addressed, can negatively impact the deployment and operation of IoT-enabled systems. This article addresses IoT security and presents an intelligent two-layer intrusion detection system for IoT. The system's intelligence is driven by machine learning techniques for intrusion detection, with the two-layer architecture handling flow-based and packet-based features. By selecting significant features, the time overhead is minimized without affecting detection accuracy. The uniqueness and novelty of the proposed system emerge from combining machine learning and selection modules for flow-based and packet-based features. The proposed intrusion detection works at the network layer, and hence, it is device and application transparent. In our experiments, the proposed system had an accuracy of 99.15% for packet-based features with a testing time of 0.357 μs. The flow-based classifier had an accuracy of 99.66% with a testing time of 0.410 μs. A comparison demonstrated that the proposed system outperformed other methods described in the literature. Thus, it is an accurate and lightweight tool for detecting intrusions in IoT systems.**

***Index Terms*—Efficiency, flow-based features, Internet of Things (IoT), intrusion detection, machine learning, packet-based features.**

I. Introduction

THE Internet of Things (IoT) has evolved to become an enabler for various applications, including healthcare, autonomous driving, education, agriculture, military, smart homes, and recently, enterprise systems. An IoT network is beneficial because it can facilitate people's interactions with real-world applications. The development of IoT systems goes hand-in-hand with advances in technologies such as sensing, actuating, cloud computing, and big data analytics and visualization [1].

An enterprise Internet of Things (E-IoT) is an anticipated extension of the IoT model. An E-IoT comprises the same IoT components and facilitates the engagement of IoT devices (e.g., sensors and actuators) to automate enterprise operations, such as business processes. An E-IoT can enhance decision-making and improve an enterprise's efficiency by collecting, aggregating, analyzing, and visualizing heterogeneous data from different sensors [2].

The development of the IoT has turned it into a mandatory enabler for many applications. However, the different components and layers of an IoT network are vulnerable to various security threats [3] that could affect whatever application the IoT is used for [4], [5]. Therefore, security is a key concern for any real-world smart environment based on the IoT model. Different approaches have been adopted to address these security threats [6]–[9].

Limitations in the computing capabilities and storage of IoT devices have made it difficult to deploy standard encryption techniques and common intrusion detection systems (IDSs). Therefore, customized intrusion detection as a second line of defense has become imperative for IoT systems. Intrusion detection classifies traffic and applications as benign or malicious [10]. Therefore, a machine learning classifier is the right tool for building an accurate IDS [11], [12]. According to the survey described in [4], a few IDSs have been specifically designed and customized for the IoT.

This study addresses the lack of an efficient IDS that is fully customized for the IoT and presents an intelligent two-layer IDS based on machine learning. The central idea of the proposed system is that it combines flow-based and packet-based features to achieve high detection accuracy. Furthermore, only the most influential features are used, which hence, reduces the detection time and the processing load for the detection process.

As discussed in Section II, previous research on intrusion detection in IoT environments has focused on either flow-based or packet-based features. Limiting the detection process to packet- or flow-specific features reduces the performance in detecting various types of attacks. Malicious actors utilize different attack techniques. These may rely on individual malicious packets, such as in port scanning, or on a group of malicious packets that can be compiled into a network flow, such as man-in-the-middle attacks. In this research, we address this gap by devising a

Manuscript received 10 April 2022; revised 15 June 2022; accepted 30 June 2022. Date of publication 18 July 2022; date of current version 8 November 2022. Paper no. TII-22-1533. *(Corresponding author: Ali Ismail Awad.)*

Mohammed M. Alani is with the School of IT Administration and Security,Seneca College of Applied Arts and Technology, Toronto, M2J 2X5, Canada, and also with Toronto Metropolitan University, Toronto, ON M5B 2H3, Canada (e-mail: m@alani.me).

Ali Ismail Awad is with the College of Information Technology, United Arab Emirates University, Al Ain 17551, UAE, with the Department of Computer Science, Electrical and Space Engineering, Luleå University of Technology, 971 87 Luleå, Sweden, also with the Electrical Engineering Department, Faculty of Engineering, Al-Azhar University, Qena 83513, Egypt, and also with the Centre for Security, Communications and Network Research, University of Plymouth, PL4 8AA Plymouth, U.K. (e-mail: ali.awad@uaeu.ac.ae; ali.awad@ltu.se).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TII.2022.3192035.

Digital Object Identifier 10.1109/TII.2022.3192035

1551-3203 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

two-layer system that detects an attack through both packet and network flow features.

A. Research Contributions

Our research into the proposed IDS makes the following contributions.

1) This research presents a novel two-layer IoT IDS based on machine learning that handles traffic at both packet level and flow level. To the best of our knowledge, the proposed two-layer architecture handling two types of features is novel and unique.
2) The proposed IDS relies on a reduced number of features, which helps to make the system more efficient and realizable. The feature selection technique used not only reduces the dimensionality of the data going into the classifier but also reduces the number of features that the system needs to collect in the data acquisition phase of deployment.
3) The selected classifiers achieve very high accuracy, low false positive rate (FPR), low false negative rate (FNR), and reduced detection time in comparison to other approaches.

Compared to other systems described in the literature, the proposed IDS for IoT applications is unique and efficient due to its use of both packet-based and flow-based features and its two-layer architecture and because it selects the most influential features. Furthermore, the proposed system works mainly at the network layer. Hence, it is application transparent and consequently, can be used as a valuable security measure for E-IoT applications.

B. Paper Layout

The rest of this article is organized as follows. The introduction is presented in Section I. Section II presents previous works relevant to this study. The information introduced in that section is later utilized in the comparative analysis of the proposed system and other systems. The architecture and components of the proposed IDS are described in Section III. The dataset used, including the flow-based and packet-based features, is discussed in Section IV. The experiments and the results are presented in Section V. Section VI is a discussion of the results and the research limitations. Finally, Section VII concludes this article, along with suggestions for future research.

II. Related Work

Intrusion detection has been the subject of a considerable amount of published research. Particular attention has been given to the deployment of different machine learning models to achieve high detection accuracy. However, intrusion detection for the IoT is still a challenge due to the data heterogeneity, limited device capabilities, and the huge number of possible applications. Therefore, an IDS for the IoT has to be a complete system [13]. Artificial intelligence, such as machine learning, has been used as a powerful tool to solve IoT security issues due to its ability to cope with data heterogeneity and velocity.

Pajouh *et al.* [14] presented a two-layer dimension reduction and two-tier classification model for anomaly-based intrusion detection in IoT backbone networks. Linear discriminant analysis (LDA) and principal component analysis (PCA) were used for feature reduction. The system uses a naive Bayes classifier to classify traffic, and then the normal traffic is passed through a certainty-factor k-nearest neighbor (CF-KNN) classifier as an additional check. The accuracy achieved was 84.86% with an FPR of 4.86% measured using the NSL-KDD dataset. However, the detection time was not reported.

Li *et al.* [15] proposed a two-stage IDS based on the Bat algorithm (BAT) and artificial intelligence for software-defined IoT networks. It is flow-based and can capture network flows to detect attacks. The system was evaluated with the KDD Cup 1999 dataset and achieved a detection accuracy of 96.42% with an FPR of 0.98%. The time overhead proportionally increased with the number of flows, and reached almost 4.5 s for 5×10^4 flows.

In [16], the authors proposed an intrusion detection framework with a dense random neural network (DnRaNN) for the IoT. It had a detection accuracy of 99.14% and 99.05% for binary class and multi-class classification, respectively. However, the time overhead and the complexity of the proposed framework were not measured.

Kamaldeep *et al.* [17] developed an IDS for the IoT cross-layer called IoT-Sentry. The system achieved an accuracy of 99.46%, evaluated on a non-standard dataset, compared to its competitors. IoT-Sentry uses only packet-based features and guards against only a limited number of attacks. No details were given of the time overhead.

Recently, in 2022, Saba *et al.* [18] proposed the deployment of deep learning via a convolutional neural network (CNN) for anomaly-based detection. This IDS can efficiently examine all the traffic flowing across an IoT system. The model was evaluated using the NID and BoT-IoT [19] datasets, attaining accuracies of 99.51% and 92.85%, respectively. The study did not consider feature optimization, and the detection time was not reported.

As demonstrated by this literature review, intrusion detection is an outstanding problem that is highly relevant to the current IoT and cybersecurity research communities. Based on a careful analysis of the relevant literature, we concluded that realizing a balance of superior detection accuracy and reduced detection time with a simple and implementable machine learning model was still a valid challenge, and this motivated the current study. Table IX briefly compares the performance of the IDSs described in the literature review with the proposed system.

III. Overview of the Proposed System

The aim of the proposed system is to provide additional protection by utilizing two different machine learning classifiers: 1) one classifier operates with packet-extracted features; 2) the other operates with flow-extracted features. Fig. 1 is an overview of the proposed system.

As shown in Fig. 1, the system operation is divided into two phases: 1) development; 2) deployment. During the development

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

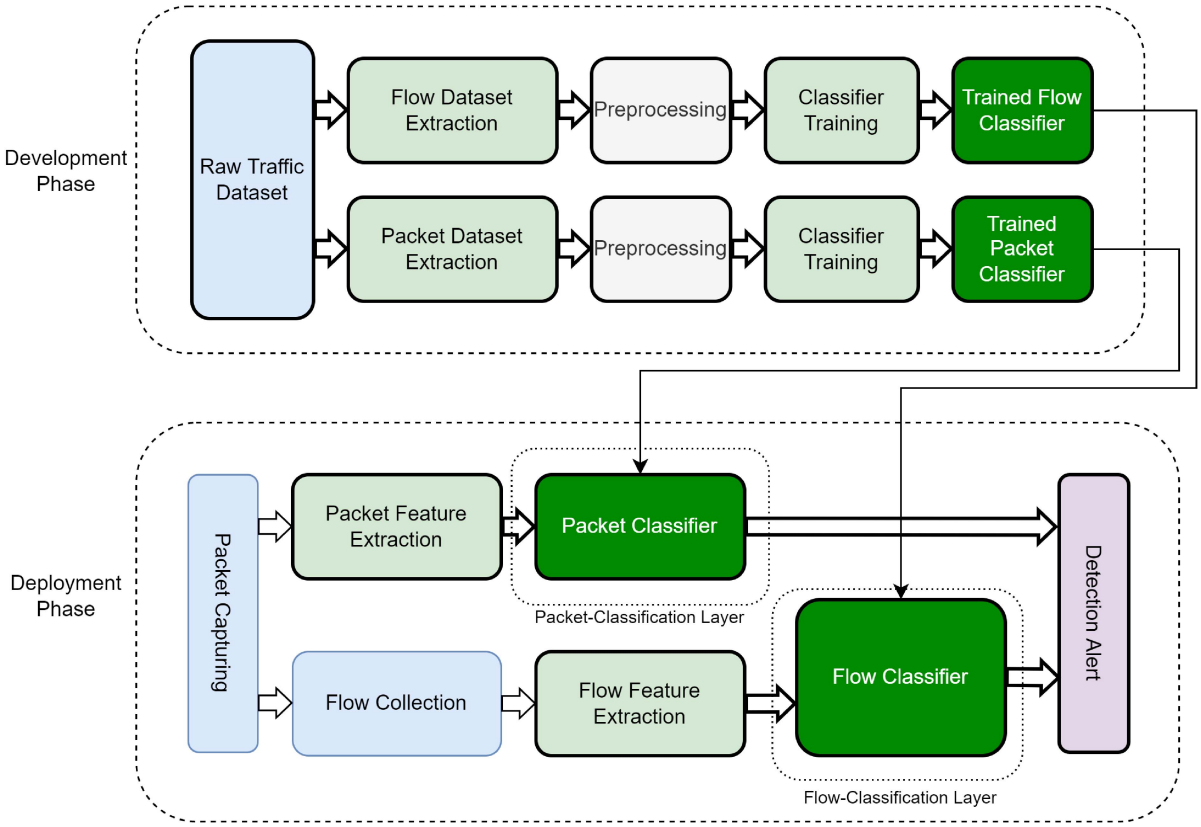


Fig. 1. Overview of the proposed IDS for the IoT. The bold blocks have been implemented in this study, while the green blocks are the core modules of the proposed system.

phase, flow and packet features are separately extracted from the original raw traffic data. Both feature datasets are preprocessed, specifically to address the observations made while individually examining them (Section IV). Then, a pipeline of classifiers is trained using each dataset independently. Once the best classifier has been identified for each dataset, it is deployed in the next phase.

In the deployment phase, all incoming and outgoing packets at a designated interface are captured. The raw traffic is then passed to two units. The first unit extracts the significant packet-level features. These are passed to the packet classifier, which assesses whether each packet is malicious or benign. The other unit collects each packet as part of a flow. When the flow is complete or the flow collection timer has expired, the flow information is passed to a flow-level feature extraction unit. From the flow features, the flow classifier decides whether a flow is malicious or benign. If a malicious packet or flow is detected, the system signals a detection alert. This alert can be used to block the source of the attack immediately.

Most of the previous works explored in Section II utilize flow-based detection, which is commonly described in the literature [20], [21]. The proposed system utilizes both packet- and flow-based features to improve the overall accuracy of detection. Some attacks, specifically reconnaissance attacks, such as port scanning and OS detection, do not use a complete communication session. For example, port scanning attacks send only one packet per port during the scanning process. This makes it difficult to identify them based on flow features, as the whole flow is just one packet. In similar attacks where flow information is scarce, packet-level detection achieves higher accuracy for a much faster detection time. These attacks are especially important to prevent because they constitute the first step of higher-risk attacks, such as botnets and remote code execution.

Another design choice we made was to use classification algorithms that do not use neural networks or deep learning. Neural networks are resource-intensive in comparison to other machine learning classifiers [22]. All blocks with thicker borders in Fig. 1 have been implemented.

IV. Dataset

As the proposed system requires data in both packet-based and flow-based formats, we selected the dataset "IoT network intrusion dataset" presented in [23]. This dataset was created by capturing raw network packets from actual IoT devices, including actual attack traffic. The dataset consists of 42 packet capture (pcap) files that include benign and attack packets. The types of attack captured in the dataset are denial-of-service (DoS) SYN-flooding, man-in-the-middle address resolution protocol (ARP) spoofing, host scanning, port scanning, and Mirai botnet attacks (host brute force, user datagram protocol (UDP) flooding, hyper-text transfer protocol (HTTP) flooding, and ACK flooding).

The dataset has 2 985 994 packets (1 229 718 malicious and 1 756 276 benign). Upon examination, we noticed that 949 284 of the packets were for the Mirai UDP flooding attack, so that

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: Packet-Based Dataset Preprocessing.

Input: Packet-based dataset (1 731 858 instances and 21 features)
Output: Balanced dataset with no missing data (1 259 982 instance and 21 features)
$Array \leftarrow RawDataset$
In $(Array)$ label-encode $icmp$ and arp features
In $(Array)$ convert $ip.flags$ and $tcp.flags$ to decimal
In $(Array)$ assign -1 to mutually exclusive features
$Array \leftarrow$
$RandomUnderSampling_MajorityClass(Array)$
$Dataset \leftarrow Array$

Algorithm 2: Flow-Based Dataset Preprocessing.

Input: Flow-based dataset (1 731 858 instances and 20 features)
Output: Balanced dataset with no absent data (1 259 982 instances and 14 features)
$Array \leftarrow RawDataset$
$Array \leftarrow Remove_HostSpecificFeatures(Array)$
In $(Array)$ label-encode
$proto, service, conn_state, history$ features
$Array \leftarrow$
$RandomOverSampling_MinorityClass(Array)$
$Dataset \leftarrow Array$

attack type was seriously imbalanced. Hence, we decided to omit two of the pcap files that hold Mirai UDP flooding attacks. This left a final raw dataset with a total of 2 151 852 packets (419 994 malicious and 1 731 858 benign).

To fulfill the goals of this research, two datasets were needed to train the two layers of the proposed system: 1) packet-based dataset; 2) flow-based dataset. The packet-based dataset would include features extracted at the packet level, which are to be used for training and testing the packet layer. On the other hand, the flow-level dataset would include flow-based features, which are to be used in training and testing the flow layer. The following sections discuss the extraction and preprocessing of these two datasets.

A. Packet-Based Dataset

Features for all packets in the 40 pcap files selected were extracted with the tshark tool [24]. The proposed system used 27 features based on the description document for the raw dataset. After extraction, all host-specific features were removed, such as the source and destination IP and media access control (MAC) addresses. This was to ensure that the trained model can generalize beyond its training dataset. A few empty features were also removed. The resulting dataset comprises 2 151 852 instances, each of which represents 21 features extracted from one packet. As the research goal is to detect any type of attack, without distinguishing the specific types, we chose to label the instances only as "malicious" or "benign." We did not use the specific attack type as a label.

Upon examining the resulting dataset, we noted the following observations.

1) The features "icmp" and "arp" are in text format.
2) The features "ip.flags" and "tcp.flags" are hexadecimal.
3) Some features have mutually exclusive values. Specifically, all transmission control protocol (TCP)-related features, such as "tcp.flags," "tcp.checksum.status," and "tcp.len," are missing in instances extracted from UDP packets, and vice versa.
4) There is a noticeable imbalance between the malicious class and the benign class, with 1 731 858 benign and 419 994 malicious instances.

To address the observations listed above, we developed the preprocessing steps shown in Algorithm 1. The first step is to perform label encoding on icmp and arp features. If a packet is an Internet control messaging protocol (ICMP) packet, the value of the icmp feature is set to 1, otherwise it is set to 0. If the packet is an ARP packet, the value of the arp feature is set to 1, otherwise it is set to 0. The second step is to convert the ip.flags and tcp.flags features from hexadecimal to decimal. For the mutually exclusive features, we set absent values to -1. We chose -1 over 0 because 0 is a valid value for these features. Hence, all UDP features will have a value of -1 for each instance of a TCP packet. Similarly, all TCP features will have a value of -1 for instances of UDP packets. The last step reduces the imbalance between the two classes by randomly undersampling benign packets, so that the final dataset contains 66.66% benign and 33.33% malicious instances. It comprises 1 259 982 instances (839 988 benign and 419 994 malicious), and all values are populated.

B. Flow-Based Dataset

The same 40 pcap files that were used to build the packet-based dataset were used for the flow-based dataset. Altogether, 20 flow features were extracted for all packets with the Zeek tool [25], and these were stored in the conn.log file. As for the packets, to ensure that the trained model can generalize beyond its training dataset, all host-specific features were removed, such as the source and destination IP addresses and flow timestamps. Two empty features (local_orig and local_resp) were also removed. The resulting dataset comprises 198 064 instances. Each instance represents 14 features extracted from one network flow. As with the packets, we labeled the instances as either "malicious" or "benign."

Again, the resulting dataset was examined, and the following observations were noted.

1) The features "proto," "service," "conn_state," and "history" are in text format.
2) The number of benign instances was 86 719, while there were 111 345 malicious instances.

To address the observations listed above, we developed the preprocessing steps shown in Algorithm 2. The first step of preprocessing is to remove host-specific features. The second step performs label encoding on the proto, service, conn_state, and history features. The final step reduces the imbalance between the two classes by randomly oversampling benign packets, so

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

TABLE I
HARDWARE AND SOFTWARE FOR THE IMPLEMENTATION ENVIRONMENT

Item	Configuration
Processor	AMD Ryzen 5 3600 4.2 GHz
RAM	128 GB
GPU	Nvidia RTX 3060Ti
Python	3.8.5
Sci-kit learn	1.0.1
Matplotlib	3.3.4

that the final dataset has 50% benign and 50% malicious instances. It comprises 222 690 instances (111 345 benign and 111 345 malicious) and 14 features. All values are populated.

V. EXPERIMENTS AND RESULTS

Table I lists the hardware and software specifications of the experimental environment.

A. Experimental Design

The experiments were implemented as follows.

1) After preprocessing both the packet-based and flow-based datasets, we split each dataset randomly, with 66.66% as a training subset and 33.33% as a testing subset.
2) We created a pipeline of five different classifiers for each dataset to be trained using the training subset. The five classifiers chosen were: 1) random forest (RF); 2) Gaussian naive Bayes (GNB); 3) xgradient boost (XGB); 4) logistic regression (LR); 5) decision tree (DT). The purpose of building this pipeline was to find the classification algorithm that would provide the best performance in comparison to the others. The pipeline was run separately for the packet features and the flow features datasets.
3) The two pipelines of classifiers were trained, one with the packet-based training subset and the other with the flow-based training subset. Testing was done with the corresponding testing subsets.
4) The best classifier in each pipeline was used to measure feature importance based on recursive feature elimination (RFE). Algorithm 3 shows the RFE steps implemented to reduce the number of features and improve efficiency [26]. Feature importance can be measured as the average decrease in impurity computed from all DTs in the forest, without assuming that the data are linearly separable [27]. This algorithm was chosen over other dimensionality reduction algorithms for two reasons: a) dimensionality reduction algorithms of a statistical nature, such as PCA and LDA, are computationally intensive, not only at the development stage but also at the deployment stage, because these algorithms require intensive preprocessing to produce the final features to be fed into the classifier; b) RFE not only reduces the number of features fed into the classifier (without any additional preprocessing) but also reduces the number of features to be captured at the data acquisition stage. This helps in improving the overall efficiency.

Algorithm 3: Recursive Feature Elimination Using Feature Importance.

Input: Dataset with n features
Output: Dataset with m features

$Array \leftarrow Dataset$
$model \leftarrow Classifier$
$TargetFeatures \leftarrow m$
while $Features(Dataset) > TargetFeatures$ **do**
  $RandomSplit(Array) \rightarrow$
   $Train_Array, Test_Array$
  train $model$ with $Train_Array$
  $importance \leftarrow FeatureImportance(model)$
  $i \leftarrow$ index of feature with lowest importance
  $Array.DeleteFeature(i)$
end
$Dataset \leftarrow Array$

5) The two pipelines of classifiers were retrained and retested using the feature-reduced datasets, with a similar 66.66% and 33.33% training/testing random split. The purpose of this stage was to measure the impact of feature selection on the performance of the classifiers.
6) To ensure that the classifiers trained using the feature-reduced dataset can generalize well beyond their training dataset, we implemented ten-fold cross-validation. The data were randomly split into ten subsets or folds. The data then went through ten cycles of training followed by testing. In each cycle, one of the ten subsets was excluded from training and used instead for testing. During the ten cycles, each of the ten subsets was used once for testing. Each cycle produced a classifier with specific performance parameters. If these parameters have a high variance, then the classifier suffers from overfitting and cannot properly generalize within the dataset. If the process produces a low variance, then the performance parameters are reliable.

B. Packet-Based Classifier Results

The five classifiers were trained using 66.66% of the packet-based dataset, as explained in Section V-A. Table II shows the results of testing using the remaining 33.33% of the packet-based dataset. XGB performed slightly better than the other classifiers, with an accuracy of 0.9937, an F_1 score of 0.9930, and a training time of 14.41 s. Hence, it was the one we chose for the RFE algorithm. Since the F_1 score falls with less than nine features, as shown in Fig. 2(a), to prevent the performance from dropping significantly, we decided to use nine features. The selected features are listed in Table III.

The next step was to test the nine-feature packet-based dataset to ensure that its performance was comparable to that of the full-feature dataset. Table IV shows that the performance metrics of XGB, RF, LR, and DT were not significantly affected, while the performance of GNB was noticeably worse.

Fig. 3 compares the performance metrics for the 21- and nine-feature datasets. For the XGB and RF classifiers, performance did not degrade with fewer features, as both accuracy and F_1

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

TABLE II
INITIAL RESULTS OF PACKET-BASED CLASSIFIER TESTING

Classifier	Accuracy	Precision	Recall	F_1 score	Training time (s)	Testing time (µs)
RF	0.9927	0.9914	0.9924	0.9919	80.81	9.7983
LR	0.8582	0.8692	0.8072	0.8274	32.98	0.0242
DT	0.9925	0.9914	0.9919	0.9916	6.27	0.1016
GNB	0.7241	0.6956	0.6355	0.6426	0.23	0.2904
XGB	0.9937	0.9919	0.9942	0.9930	14.41	0.3911

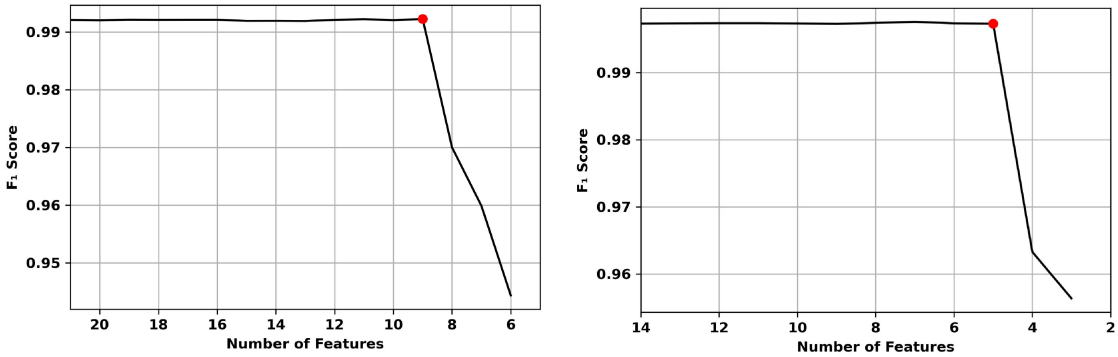


Fig. 2. F_1 score plotted against the number of features. (a) Number of packet-based features. (b) Number of flow-based features.

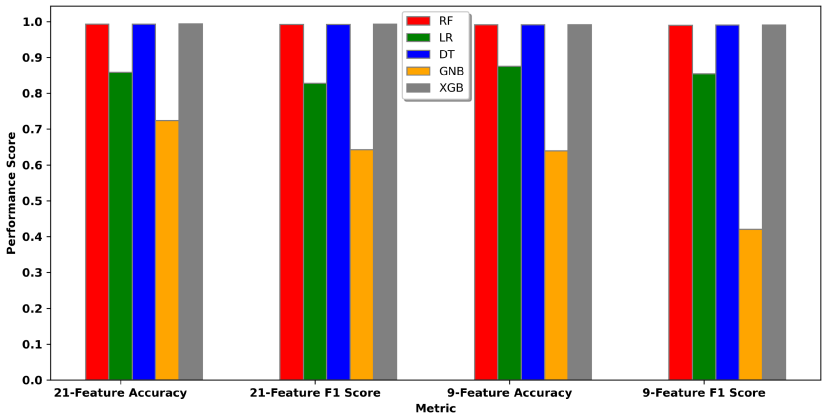


Fig. 3. Comparison of accuracy and F_1 score for the classifiers with 21- and nine-feature packet-based datasets.

TABLE III
FEATURES USED FROM PACKET-BASED AND FLOW-BASED DATASETS

Packet-based	Flow-based
frame.len, ip.len	id.orig_p
ip.ttl, ip.flags	id.resp_p
arp, tcp.window_size	duration
tcp.dstport, tcp.srcport	conn_state
udp.dstport	history

score remained above 99%. Furthermore, Fig. 4 shows that the training and testing times witnessed a noticeable improvement compared to the 21-feature dataset.

For the ten-fold cross-validation, Table V shows the means and standard deviations of the accuracy and F_1 score for the five classifiers when applied to the nine-feature packet-based dataset. As the table shows, the performance of XGB, RF, and DT are consistent with the previously obtained results. In addition, all the classifiers had a low standard deviation. This proves that the classifiers can generalize well beyond their training dataset.

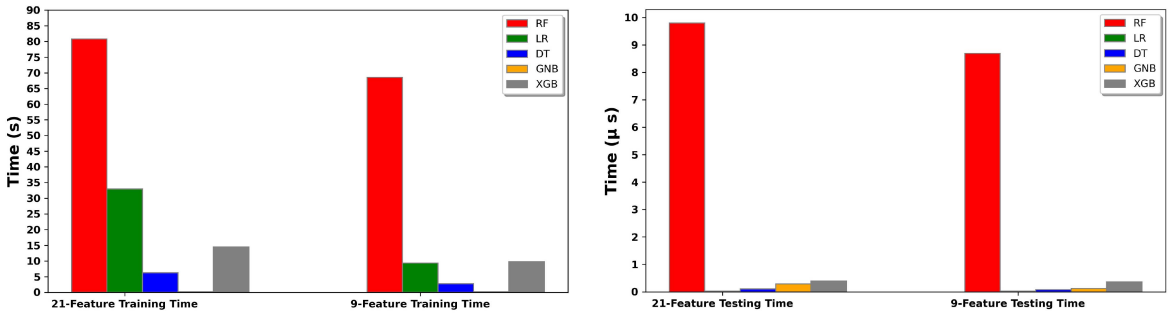
C. Flow-Based Classifier Results

The selected five classifiers were trained using 66.66% of the flow-based dataset, as explained in Section V-A. Table VI shows the results of testing using the remaining 33.33% of the dataset. RF performed slightly better than the other classifiers, with an accuracy of 0.9973, an F_1 score of 0.9974, and a training time of 7.88 s. Therefore, we chose it for the RFE algorithm to select a low number of features to improve the efficiency of

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RESULTS OF NINE-FEATURE PACKET-BASED CLASSIFIER TESTING

Classifier	Accuracy	Precision	Recall	F_1 score	Training time (s)	Testing time (µs)
RF	0.9910	0.9887	0.9914	0.9900	68.526	8.6922
LR	0.8755	0.8724	0.8419	0.8542	9.371	0.0240
DT	0.9912	0.9890	0.9914	0.9902	2.777	0.0772
GNB	0.6392	0.4483	0.4883	0.4205	0.162	0.1257
XGB	0.9915	0.9890	0.9921	0.9905	9.753	0.3571



(a) (b)

Fig. 4. Time overhead measured for training and testing processes for 21- and nine-feature packet-based datasets. (a) Training times. (b) Testing times.

TABLE V
RESULTS OF TEN-FOLD CROSS-VALIDATION OF THE NINE-FEATURE PACKET-BASED DATASET

	RF		LR		DT		GNB		XGB	
	Accuracy	F_1 score	Accuracy	F_1 score	Accuracy	F_1 score	Accuracy	F_1 score	Accuracy	F_1 score
Mean	0.991244	0.986926	0.808100	0.620940	0.991426	0.987190	0.638323	0.064653	0.991788	0.987758
Std Dev	0.000202	0.000306	0.086380	0.250322	0.000248	0.000370	0.001011	0.002034	0.000223	0.000336

TABLE VI
INITIAL RESULTS OF FLOW-BASED CLASSIFIER TESTING

Classifier	Accuracy	Precision	Recall	F_1 score	Training time (s)	Testing time (µs)
RF	0.9973	0.9974	0.9974	0.9974	7.88	8.0911
LR	0.5663	0.5711	0.5667	0.5596	3.31	0.0001
DT	0.9965	0.9966	0.9966	0.9966	0.35	0.0273
GNB	0.5040	0.6872	0.5029	0.3410	0.03	0.2460
XGB	0.9969	0.9970	0.9970	0.9970	2.59	0.6838

TABLE VII
RESULTS OF FIVE-FEATURE FLOW-BASED CLASSIFIER TESTING

Classifier	Accuracy	Precision	Recall	F_1 score	Training time (s)	Testing time (µs)
RF	0.9971	0.9972	0.9972	0.9972	8.35	7.8034
LR	0.9479	0.9498	0.9479	0.9479	0.31	0.0001
DT	0.9962	0.9963	0.9963	0.9963	0.23	0.1372
GNB	0.8385	0.8457	0.8387	0.8378	0.01	0.1366
XGB	0.9966	0.9967	0.9967	0.9967	2.17	0.4102

the system. Since the F_1 score falls with less than five features, as shown in Fig. 2(b), to prevent the performance from dropping significantly, we decided to use five features, as listed in Table III.

We then tested the five-feature flow-based dataset to ensure that its performance was comparable to that of the full-feature dataset. Table VII shows that the performance metrics of XGB, RF, and DT were not significantly affected, while the performance of LR and GNB was noticeably better.

Fig. 5 compares the performance metrics for the 14- and five-feature datasets. For the XGB and RF classifiers, performance did not degrade with fewer features, as the accuracy and F_1

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

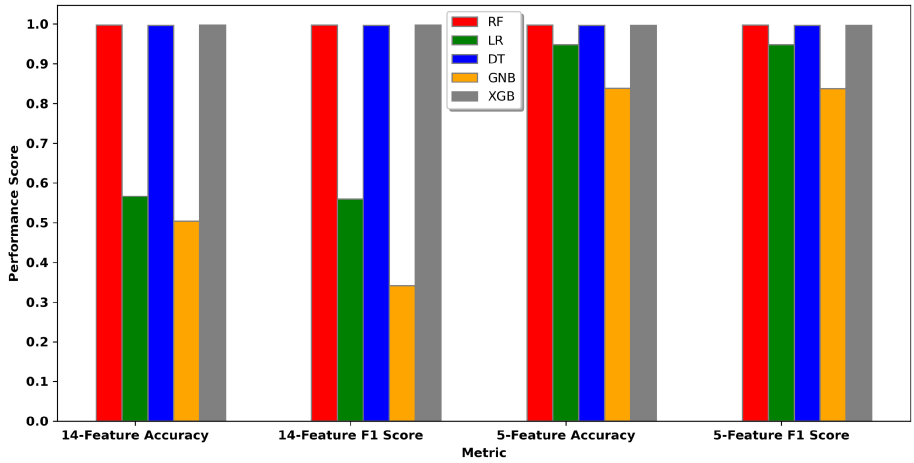
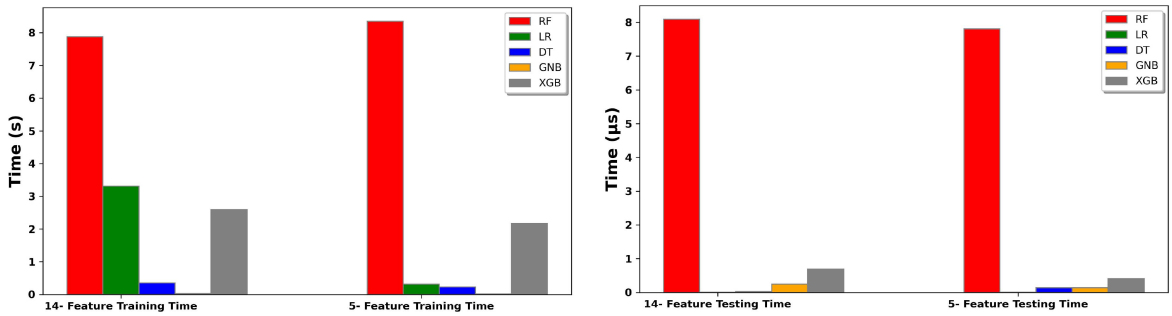


Fig. 5. Comparison of accuracy and F_1 score for the classifiers with 14- and five-feature flow-based datasets.



(a) (b)

Fig. 6. Time overhead measured for training and testing processes for 14- and five-feature flow-based datasets. (a) Training times. (b) Testing times.

TABLE VIII
RESULTS OF TEN-FOLD CROSS-VALIDATION OF THE FIVE-FEATURE FLOW-BASED DATASET

	RF		LR		DT		GNB		XGB	
	Accuracy	F_1 score	Accuracy	F_1 score	Accuracy	F_1 score	Accuracy	F_1 score	Accuracy	F_1 score
Mean	0.997198	0.997195	0.948332	0.949864	0.996542	0.996541	0.837316	0.824360	0.996946	0.996945
Std Dev	0.000287	0.000283	0.001276	0.001283	0.000343	0.000336	0.001157	0.001900	0.000173	0.000169

score remained above 99%. Furthermore, Fig. 6 indicates that the training and testing times were significantly shorter compared to the 14-feature dataset.

For the ten-fold cross-validation, Table VIII shows the mean and standard deviation of the accuracy and F_1 score for the five classifiers when applied to the five-feature flow-based dataset. As shown in the table, the performance of XGB, RF, and DT is consistent with the results previously obtained. In addition, all the classifiers had a low standard deviation. This proves that these classifiers can generalize well beyond the training dataset.

D. Classifier Selection

From the results of the classifier pipeline for the nine-feature packet-based dataset (Table IV) and the five-feature flow-based dataset (Table VII), notice that the difference in accuracy and F_1 score between RF and XGB is minimal. However, as one of our research goals is to produce an efficient system, we consequently used XGB, as it has the best combination of high accuracy and low testing time.

For this classifier, the measured FPR was 0.96% for the packet-based classifier (Table IX), whereas the FNR was 0.61%. The flow-based classifier had a lower FPR of 0.26% and a lower FNR of 0.40%. These low rates indicate that, in comparison to previous approaches, both classifiers performed very well with a low number of features extracted from captured traffic.

Figs. 7 and 8 show the confusion matrix plots for the XGB classifier using nine-feature packet-based dataset and five-feature flow-based dataset, respectively.

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
COMPARISON OF THE PROPOSED SYSTEM WITH STATE-OF-THE-ART APPROACHES

Literature	Performance			Time overhead		Model
	Accuracy (%)	FPR (%)	F_1 score	Training	Testing	
[14] (—)	84.86	4.86	–	–	–	CF-KNN, LDA, and PCA
[15] (Flow-based)	96.42	0.98	–	–	4.5 s	BAT and RF
[16] (—)	99.14	–	0.992	–	–	DnRaNN
[17] (Packet-based)	99.46	–	0.992	–	–	Ensemble learning, RF
[18] (Flow-based)	92.85	–	–	–	–	Deep learning (CNN)
Proposed system						
Packet-based	99.15	0.96	0.990	9.753 s	0.357 µs	RF, LR, DT, GNB, and XGB
Flow-based	99.66	0.26	0.996	2.16 s	0.410 µs	RF, LR, DT, GNB, and XGB

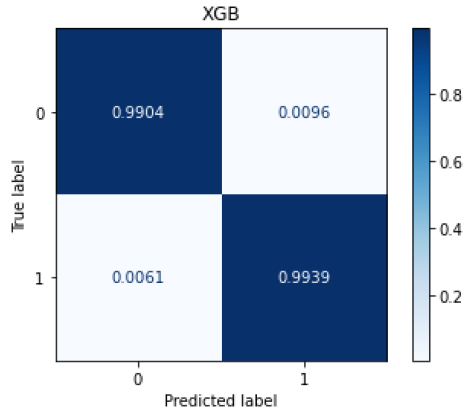


Fig. 7. Confusion matrix for XGB classifier using the packet-based nine features.

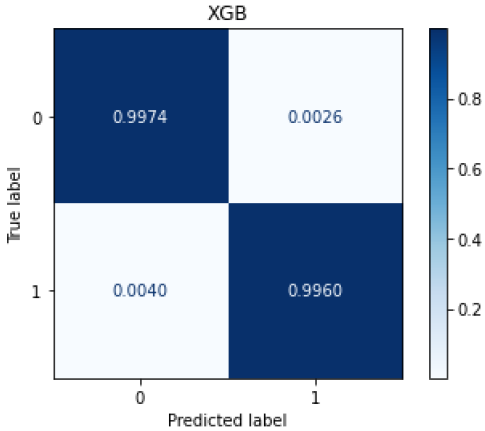


Fig. 8. Confusion matrix for XGB classifier using the flow-based five features.

VI. DISCUSSION AND COMPARATIVE ANALYSIS

Many research studies have tried addressing intrusion detection in IoT devices. These studies, as we explained in Section II, have mostly focused on high accuracy while ignoring barriers to deployment in real-world applications. This study presents a novel two-layer solution that is designed to harvest the benefits of packet- and flow-level features. Using only the features selected by RFE produced a noticeable reduction in training time. The advantage of using RFE over other dimensionality reduction solutions is that fewer features need to be captured and extracted in the data acquisition phase of a real-time deployment. This improves efficiency and reduces processing overhead, which are important for the resource-constrained environment of the IoT.

Table IX compares the performance metrics for our proposed system with those for related approaches. The table confirms that our proposed system outperforms related approaches in terms of accuracy and testing time. In addition, our proposed system delivers a noticeable reduction in the FPR and FNR. At the packet level, the FPR was 0.96%, and the FNR was 0.61%, whereas at the flow-level, the FPR was only 0.26% and the FNR only 0.40%.

As Section II suggests, most previous works rely on flow-based features to detect intrusion. That works well if the attacker exchanges a reasonable amount of information with the target. However, other types of attack, such as port scanning, ping sweep, and OS fingerprinting, are better detected using packet-based features. This is because the attacker targets different ports using different packets, which makes it impossible to follow a network flow. Hence, the combination of flow-based and packet-based features used in our system ensures that different types of attacks can be detected.

Another advantage of our proposed system over other systems is that it utilizes XGB as a classification algorithm. This algorithm requires noticeably less intensive processing in comparison to other deep learning algorithms [22]. XGB was chosen not only for its high performance but also due to the processing limitations of the deployment environment.

VII. CONCLUSION

In this article, we presented a novel two-layer IoT IDS based on machine learning. The proposed system examines features extracted from packets as well as from network flows. Combining these two defenses provides the maximum possible protection. In testing, the proposed system had a very high accuracy of 99.15% at the packet level and 99.66% at the flow level. This high accuracy was combined with noticeable efficiency improvements, namely, detection times of 0.357 μs at the packet level and 0.410 μs at the flow level. For the packet-based classifier, the FPR was only 0.96% and the FNR was 0.61%. For the flow-based classifier, the FPR was only 0.26%, while the FNR was 0.40%. These metrics show that the proposed system outperforms previous approaches. Our future work will focus on the decentralized implementation of the proposed two-layer system in the edge to offload some of the processing from IoT devices.

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] B. Ali and A. I. Awad, "Cyber and physical security vulnerability assessment for IoT-based smart homes," *Sensors*, vol. 18, no. 3, 2018, Art. no. 817. [Online]. Available: https://doi.org/10.3390/s18030817

[2] S. Ahmad and D. Kim, "A multi-device multi-tasks management and orchestration architecture for the design of enterprise IoT applications," *Future Gener. Comput. Syst.*, vol. 106, pp. 482–500, 2020. [Online]. Available: https://doi.org/10.1016/j.future.2019.11.030

[3] M. Mamdouh, A. I. Awad, A. A. Khalaf, and H. F. Hamed, "Authentication and identity management of IoHT devices: Achievements, challenges, and future directions," *Comput. Secur.*, vol. 111, 2021, Art. no. 102491. [Online]. Available: https://doi.org/10.1016/j.cose.2021.102491

[4] M. F. Elrawy, A. I. Awad, and H. F. Hamed, "Intrusion detection systems for IoT-based smart environments: A survey," *J. Cloud Comput.*, vol. 7, no. 1, pp. 1–20, 2018. [Online]. Available: https://doi.org/10.1186/s13677-018-0123-6

[5] A. I. Awad and J. Abawajy, *Security and Privacy in the Internet of Things: Architectures, Techniques, and Applications*, 1st ed. Hoboken, NJ, USA: Wiley, 2021.

[6] M. Hassaballah, M. A. Hameed, A. I. Awad, and K. Muhammad, "A novel image steganography method for industrial Internet of Things security," *IEEE Trans. Ind. Informat.*, vol. 17, no. 11, pp. 7743–7751, Nov. 2021. [Online]. Available: https://doi.org/10.1109/TII.2021.3053595

[7] Q.-D. Ngo, H.-T. Nguyen, V.-H. Le, and D.-H. Nguyen, "A survey of IoT malware and detection methods based on static features," *ICT Exp.*, vol. 6, no. 4, pp. 280–286, 2020. [Online]. Available: https://doi.org/10.1016/j.icte.2020.04.005

[8] M. M. Alani, "Detection of reconnaissance attacks on IoT devices using deep neural networks," in *Advances in Nature-Inspired Cyber Security and Resilience*, S. K. Shandilya, N. Wagner, V. Gupta, and A. K. Nagar, Eds., Cham, Switzerland:Springer International Publishing, 2022, pp. 9–27.

[9] A. A. Cook, G. Mısırlı, and Z. Fan, "Anomaly detection for IoT time-series data: A survey," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6481–6494, Jul. 2020. [Online]. Available: https://doi.org/10.1109/JIOT.2019.2958185

[10] N. Lu, D. Li, W. Shi, P. Vijayakumar, F. Piccialli, and V. Chang, "An efficient combined deep neural network based malware detection framework in 5G environment," *Comput. Netw.*, vol. 189, 2021, Art. no. 107932. [Online]. Available: https://doi.org/10.1016/j.comnet.2021.107932

[11] P. Illy, G. Kaddoum, K. Kaur, and S. Garg, "ML-based IDPS enhancement with complementary features for home IoT networks," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 2, pp. 772–783, Jun. 2022. [Online]. Available: https://doi.org/10.1109/TNSM.2022.3141942

[12] M. M. Alani and A. I. Awad, "AdStop: Efficient flow-based mobile adware detection using machine learning," *Comput. Secur.*, vol. 117, 2022, Art. no. 102718. [Online]. Available: https://doi.org/10.1016/j.cose.2022.102718

[13] E. Benkhelifa, T. Welsh, and W. Hamouda, "A critical review of practices and challenges in intrusion detection systems for IoT: Toward universal and resilient systems," *IEEE Commun. Surv. Tut.*, vol. 20, no. 4, pp. 3496–3509, Oct.–Dec. 2018. [Online]. Available: https://doi.org/10.1109/COMST.2018.2844742

[14] H. H. Pajouh, R. Javidan, R. Khayami, A. Dehghantanha, and K.-K. R. Choo, "A two-layer dimension reduction and two-tier classification model for anomaly-based intrusion detection in IoT backbone networks," *IEEE Trans. Emerg. Topics Comput.*, vol. 7, no. 2, pp. 314–323, Apr.–Jun. 2019. [Online]. Available: https://doi.org/10.1109/TETC.2016.2633228

[15] J. Li, Z. Zhao, R. Li, and H. Zhang, "AI-based two-stage intrusion detection for software defined IoT networks," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 2093–2102, Apr. 2019. [Online]. Available: https://doi.org/10.1109/JIOT.2018.2883344

[16] S. Latif et al., "Intrusion detection framework for the Internet of Things using a dense random neural network," *IEEE Trans. Ind. Informat.*, vol. 18, no. 9, pp. 6435–6444, Sep. 2022. [Online]. Available: https://doi.org/10.1109/TII.2021.3130248

[17] Kamaldeep, M. Malik, M. Dutta, and J. Granjal, "IoT-Sentry: A cross-layer-based intrusion detection system in standardized Internet of Things," *IEEE Sensors J.*, vol. 21, no. 24, pp. 28066–28076, Dec. 2021. [Online]. Available: https://doi.org/10.1109/JSEN.2021.3124886

[18] T. Saba, A. Rehman, T. Sadad, H. Kolivand, and S. A. Bahaj, "Anomaly-based intrusion detection system for IoT networks through deep learning model," *Comput. Elect. Eng.*, vol. 99, 2022, Art. no. 107810. [Online]. Available: https://doi.org/10.1016/j.compeleceng.2022.107810

[19] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: BoT-IoT dataset," *Future Gener. Comput. Syst.*, vol. 100, pp. 779–796, 2019. [Online]. Available: https://doi.org/10.1016/j.future.2019.05.041

[20] A. Sperotto, G. Schaffrath, R. Sadre, C. Morariu, A. Pras, and B. Stiller, "An overview of IP flow-based intrusion detection," *IEEE Commun. Surv. Tut.*, vol. 12, no. 3, pp. 343–356, Jul.–Sep. 2010. [Online]. Available: https://doi.org/10.1109/SURV.2010.032210.00054

[21] A. F. AlEroud and G. Karabatis, "Queryable semantics to detect cyber-attacks: A flow-based detection approach," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 48, no. 2, pp. 207–223, Feb. 2018. [Online]. Available: https://doi.org/10.1109/TSMC.2016.2600405

[22] A. Géron, *Hands-On Machine Learning With Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*. Sebastopol, CA, USA: O'Reilly Media, Inc., 2019.

[23] H. Kang, D. H. Ahn, G. M. Lee, J. D. Yoo, K. H. Park, and H. K. Kim, "IoT network intrusion dataset," 2019. [Online]. Available: https://doi.org/10.21227/q70p-q449

[24] *tshark(1)*. Mar. 2022, Accessed: Mar. 9, 2022. [Online]. Available: https://www.wireshark.org/docs/man-pages/tshark.html

[25] The zeek network security monitor. Mar. 2022, Accessed: Mar. 30, 2022. [Online]. Available: https://zeek.org

[26] M. M. Alani, "Implementation-oriented feature selection in UNSW-NB15 intrusion detection dataset," in *Proc. Int. Conf. Intell. Syst. Des. Appl.*, 2022, pp. 548–558.

[27] S. Raschka, Y. Liu, and V. Mirjalili, *Mach. Learn. With PyTorch and Scikit-Learn*. Birmingham, U.K.: Packt Publishing, 2022.



Mohammed M. Alani (Senior Member, IEEE) received the Ph.D. in computer engineering from College of Engineering, Nahrain University, Iraq, in 2007, with a specialization in network security.

He was a Professor and a Cybersecurity Expert in many countries around the world. His experience includes serving as VP of Academic Affairs in the United Arab Emirates, network and security consultancies in the Middle East, and Cybersecurity Program Manager in Toronto Canada. He is currently a Cybersecurity Professor with Seneca College, Toronto, ON, Canada, and a Research Fellow with Ryerson University, Toronto. He has authored or coauthored four books in different areas of networking and cybersecurity along with many research papers published in highly ranked journals and conferences. His current research interests include applications of ML in cybersecurity and ML security.

Dr. Alani is a Senior Member of the Association for Computing Machinery (ACM).



Ali Ismail Awad (Senior Member, IEEE) received the Ph.D. degree in computer science from Kyushu University, Fukuoka, Japan, in 2012. He is currently an Associate Professor with the College of Information Technology (CIT), United Arab Emirates University (UAEU), Al Ain, United Arab Emirates. He is also an Associate Professor (Docent) with the Department of Computer Science, Electrical and Space Engineering, Luleå University of Technology, Luleå, Sweden, where he also served as a coordinator of the Master Program in Information Security from 2017 to 2020. He is an Associate Professor with the Electrical Engineering Department, Faculty of Engineering, Al-Azhar University at Qena, Qena, Egypt. He is also a Visiting Researcher with the University of Plymouth, Plymouth, U.K.

He has edited or coedited eight books and authored or coauthored several journal articles and conference papers in these areas. His research interests include cybersecurity, network security, Internet of Things security, and image analysis with biometrics and medical imaging applications.

Dr. Awad is an Editorial Board Member of the *Future Generation Computer Systems Journal*, *Computers and Security Journal*, *the Internet of Things*, *Engineering Cyber Physical Human Systems Journal*, and *Health Information Science and Systems Journal*.

Authorized licensed use limited to: Alliance University. Downloaded on October 14,2026 at 07:23:48 UTC from IEEE Xplore. Restrictions apply.